Geomagnetic Activity Suppresses Human Violence and Alcohol Consumption

Evidence from 13,142 Days of Conflict Data

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github.com/kirillchekanov/universe-cell

Abstract

We report a robust statistical association between geomagnetic activity (Kp index) and organized human violence, using 463,621 conflict incidents from UCDP GED 25.1 (1989–2024). During geomagnetic storms ($Kp \geq 5$), daily conflict incidents decrease by 23% globally ($p<0.001$, $N=13{,}142$ days), rising to 49% over 14-day windows and 83% in the Middle East. The effect strengthens at higher latitudes and in winter, consistent with a serotonin/melatonin mediation pathway. Supporting evidence: solar minimum predicts epidemics ($r=-0.547$, $p=0.001$) and economic crises ($r=-0.376$, $p=0.026$); solar activity predicts conflicts via Granger causality ($p=0.012$, lag 2yr); solar activity negatively correlates with alcohol consumption across Scandinavia ($r=-0.61$ to -0.77, $p<0.01$), confirmed weekly via CDC surveillance ($r=-0.135$, $p=0.023$, $N=286$ weeks). Latitude gradient of the alcohol-solar effect ($r=-0.515$, $p=0.049$, $N=18$ countries) supports a geomagnetic rather than direct UV mechanism.

1. Introduction

Chizhevsky (1926) first noted correlations between sunspot cycles and historical conflict. Prior work suffered from small samples and inadequate controls for temporal autocorrelation. We address these limitations using UCDP GED 25.1 ($N=463{,}621$ incidents), Kp index from GFZ Potsdam ($N=34{,}471$ days, 1932–2026), and multiple validation strategies including permutation tests ($n=3{,}000$), AR(1) surrogate tests, and Granger causality.

2. Data

- **Conflict:** UCDP GED 25.1, 463,621 georeferenced events, 1989–2024
- **Geomagnetic:** Kp index, GFZ Potsdam, daily means, 1932–2026
- **Solar:** International Sunspot Number v2.0, SILSO, 1700–2025
- **Alcohol:** World Bank SH.ALC.PCAP.LI, 18 countries, 2000–2020
- **CDC:** Weekly Excessive Drinking surveillance, USA, 2019–2024 (N=286 weeks)

3. Results

3.1 Geomagnetic Storms and Organized Violence

Daily conflict incidents decrease monotonically with increasing Kp, from 35.5 events/day at Kp 0–1 to 20.2 events/day at Kp 7–10, analogous to a pharmacological dose-response curve. All results survive permutation testing (p_perm<0.001) and AR(1) surrogate tests.

Condition	Storm (Kp≥5)	Quiet (Kp<2)	Reduction	p
All days	24.97/day	32.57/day	–23.3%	<0.001
Winter	22.31/day	31.87/day	–30.0%	<0.001
14-day window	16.70/day	32.75/day	–49.1%	<0.001
Middle East, 14d	2.14/day	12.43/day	–82.8%	<0.001

Violence type heterogeneity: State-based (type 1) and non-state (type 2) violence decrease significantly (r=–0.092, r=–0.091). One-sided violence (type 3) shows the opposite direction, consistent with two distinct neurobiological pathways.

Latitude gradient: Tropics 0–20°: –8.0% (ns); Subtropics 20–40°: –20.8% (p<0.001); Scandinavia 55–65°: –30.0% (p<0.001). Consistent with geomagnetic field intensity increasing toward poles.

3.2 Solar Minimum as Immune Deficiency

- More epidemic outbreaks: r=–0.547, p=0.001, N=78 years
- More economic crises: r=–0.376, p=0.026
- Granger causality SN→conflicts: F=4.57, p=0.012, lag 2 years

3.3 Geomagnetic Activity and Alcohol Consumption

Annual solar activity negatively predicts national alcohol consumption across northern European countries, with effect size increasing with latitude (r=–0.515, p=0.049, N=18 countries). At weekly timescales, CDC Excessive Drinking surveillance (USA, 2019–2024) correlates negatively with Kp (r=–0.135, p=0.023, N=286 weeks).

Country	Latitude	r	p
Iceland	65°N	–0.768	<0.001
Norway	60°N	–0.686	0.001
Denmark	56°N	–0.656	0.001
United Kingdom	53°N	–0.624	0.003
Finland	64°N	–0.610	0.003
Sweden	59°N	+0.076	0.74 (ns)

Notable exception: Sweden (r=+0.076, ns) operates Systembolaget — a state alcohol retail monopoly — suggesting institutional constraints can override neurobiological signals.

3.4 Proposed Mechanism

Geomagnetic activity affects pineal gland melatonin synthesis via magnetoreception (Lerchl et al. 1998, demonstrated in vitro). We propose two downstream pathways:

Pathway 1 (Serotonin): Kp↑ → serotonin↑ → impulsivity↓ → organized violence↓ [confirmed]; alcohol consumption↓ [confirmed]; depression↓ [literature]

Pathway 2 (Neural excitability): Kp↑ → cortical excitability↑ → psychosis↑ [literature]; one-sided violence↑ [weak signal]

4. Robustness Checks

Test	Result
Permutation test (n=3,000)	p<0.001 for primary finding
AR(1) surrogate test	p<0.001
Granger War→Kp (reverse)	p=0.87 (no reverse causality)
Walk-forward CV 2023–2024	MAE=7.8%
Planetary first-difference	Kp survives; Jupiter/Saturn fail
Lunar cycle	p=0.49 (null result)

5. Limitations

- Annual alcohol data: only 21 years per country; wide confidence intervals
- Conflict reporting bias: UCDP coverage improves over time; residual trend may persist
- Mechanism: serotonin pathway hypothesized; direct in vivo validation lacking
- One-sided violence paradox: opposite direction requires independent replication
- Sweden anomaly: institutional confounders may mask biological signals

6. Discussion

The Kp-violence association survives all statistical tests applied to 13,142 days of data. The dose-response relationship, latitude gradient, seasonal modulation, and psychiatric literature correspondence (meta-analysis r=–0.171, p<0.001, N=911 patients, 9 studies) collectively support a neurobiological mechanism over confounding.

Conflict forecasting: The 14-day accumulation effect suggests NOAA 2-week geomagnetic forecasts could serve as inputs to conflict early-warning systems, particularly for the Middle East (–83%).

Public health: Solar cycle downturn (2026–2028 predicted) correlates historically with increased alcohol consumption, epidemics, and economic crises. Proactive interventions during solar minima may reduce associated burden.

7. Conclusion

Geomagnetic activity suppresses organized human violence by 23–83% depending on region and accumulation window. The effect is robust to multiple statistical controls, survives replication across conflict types and geographic regions, and is consistent with a serotonin/melatonin mediation pathway supported by alcohol consumption data across 18 countries at both annual and weekly timescales. Solar minimum periods independently predict epidemic and economic crisis timing.

These findings are correlational; causal mechanism requires experimental validation. The effect size, consistency, and biological plausibility warrant attention from conflict researchers, public health officials, and space weather scientists.

Data and Code

All data and analysis code: **github.com/kirillchekanov/universe-cell**
Sources: UCDP GED 25.1 · GFZ Potsdam Kp · SILSO SSN · World Bank API · CDC surveillance

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